

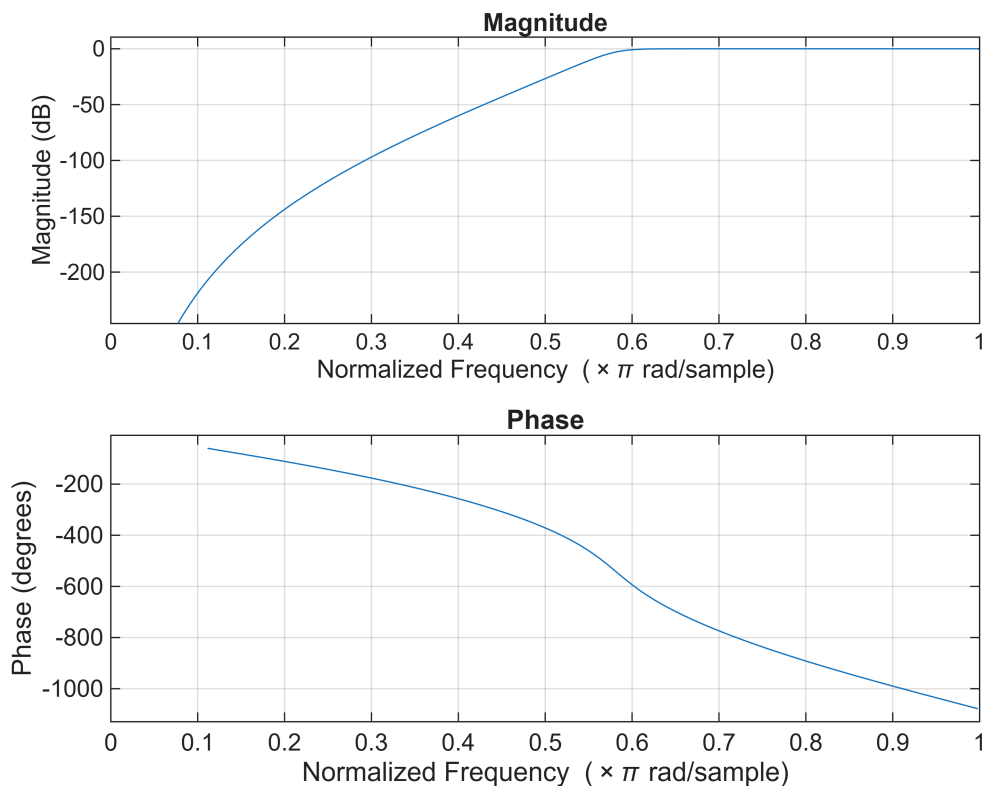
```

%% EX NO: 1 Butterworth IIR Filters
% 1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using
MATLAB
%
% Objectives:
% 1. To determine the minimum filter order and cutoff frequency required
%    to satisfy the design constraints.
% 2. To design and analyze the magnitude response, phase response
%    characteristics of the Butterworth IIR High-pass filter for the
%    given passband and stopband specifications.
% 3. To analyze the impulse response, step response and group delay
%    characteristics of the Butterworth IIR High-pass filter.
% 4. To examine the pole-zero plot and verify the stability and
%    performance of the designed High-pass filter.

clc; clear
Fs=10000; Fp=3000; Fst=2000;
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a]=butter(N,Wn,'high');

% Magnitude & Phase
freqz(b,a)

```

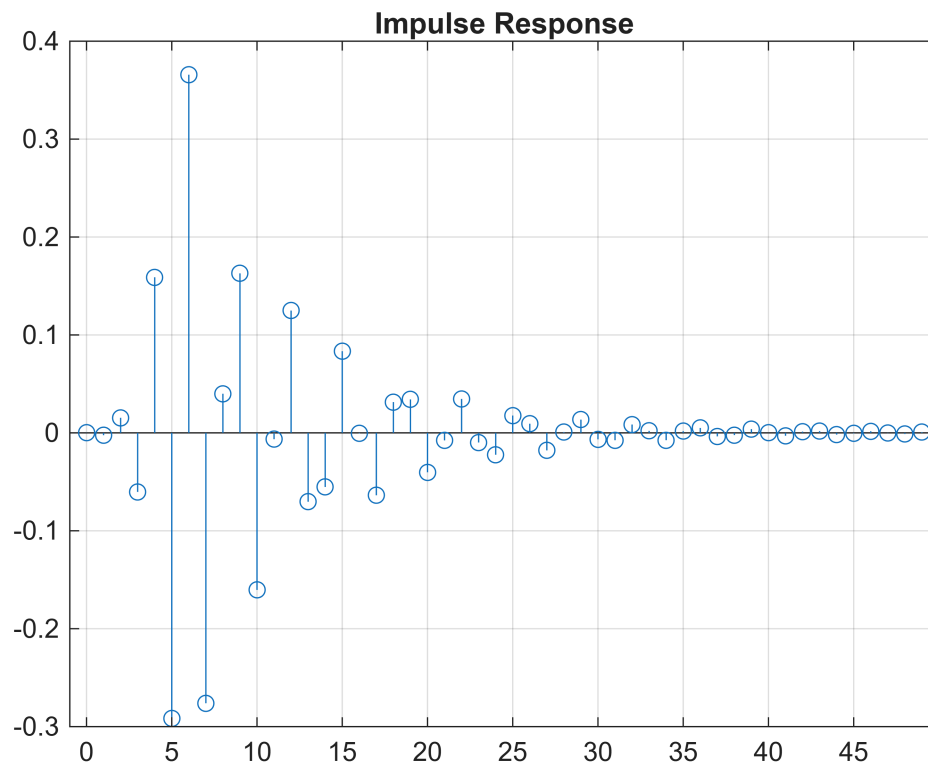


```

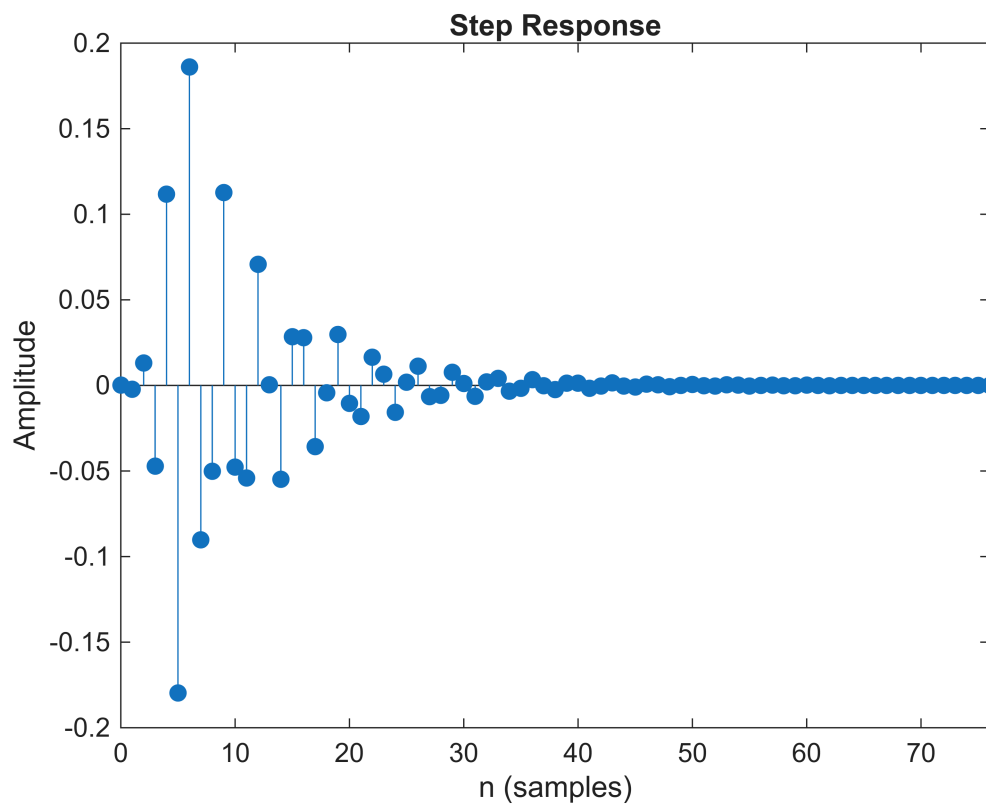
% Impulse Response
figure

```

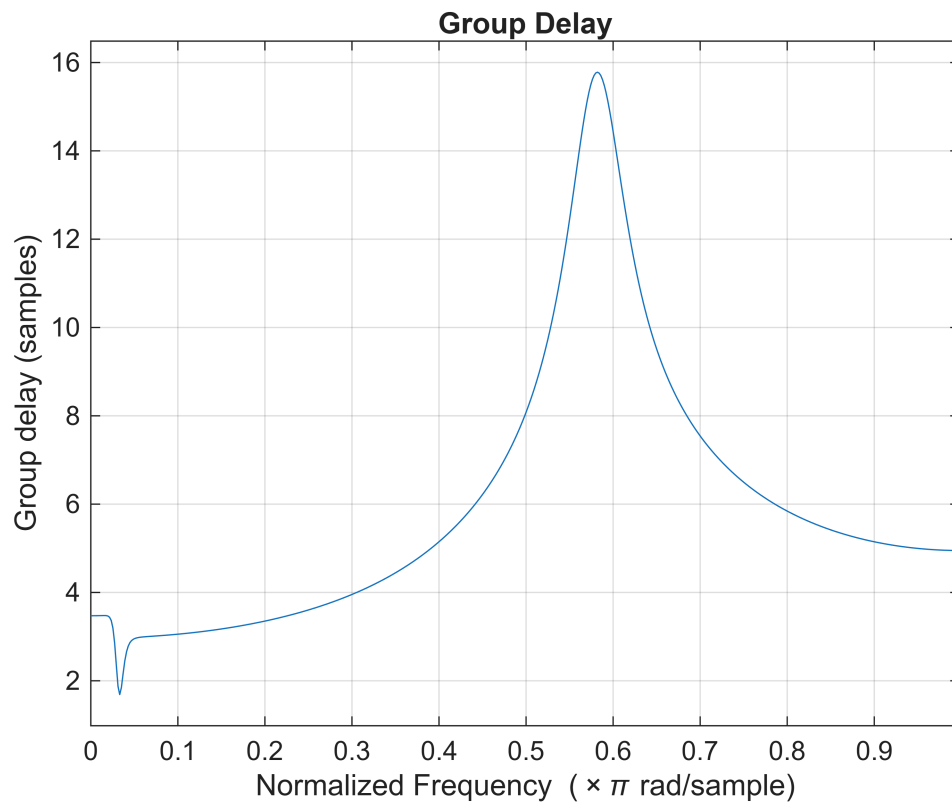
```
[h,n]=impz(b,a,50);
stem(n,h), grid on
title('Impulse Response')
```



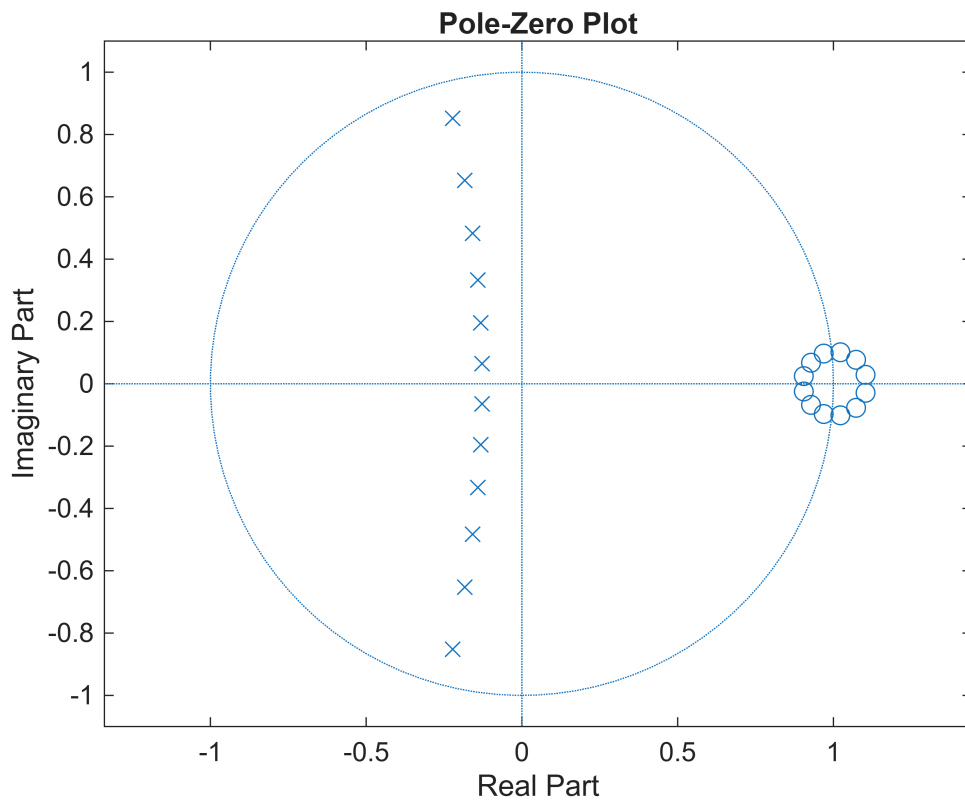
```
% Step Response
figure
stepz(b,a)
title('Step Response')
```



```
% Group Delay  
figure  
grpdelay(b,a)  
title('Group Delay')
```



```
% Pole-Zero Plot  
figure  
zplane(b,a)  
title('Pole-Zero Plot')
```



```
fprintf('Minimum Filter Order (N) = %d\n',N);
```

Minimum Filter Order (N) = 12

```
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

Cutoff Frequency (Wn) = 0.5807